

GraphRAG for Biomedical Literature

Knowledge-Graph Augmented Retrieval over Breast Cancer Research

Project Proposal

Domain	Biomedical NLP - Breast Cancer Literature
Corpus	PubMed abstracts and full-text articles
System type	Graph + Vector Hybrid Retrieval (GraphRAG)
NER approach	scispaCy + BiomedBERT + custom patterns
Entity linking	UMLS CUI resolution via scispaCy EntityLinker
Embedding model	BGE-base-en-v1.5 or BiomedBERT (dim=768)
Vector store	FAISS IndexFlatIP
Graph store	CSV triples (extensible to Neo4j AuraDB)
Status	Proposal - awaiting approval

Contents

1. Executive Summary

Biomedical research on breast cancer is vast, fast-moving, and distributed across tens of thousands of publications. Researchers and clinicians need to quickly answer questions such as: What therapies target breast cancer stem cells? Which biomarkers predict resistance to treatment? What nanoparticle delivery systems have demonstrated efficacy in breast cancer models?

This proposal describes a Graph-augmented Retrieval system (GraphRAG) designed to operate over a curated corpus of biomedical literature. The system combines two complementary retrieval strategies: dense semantic search over encoded text chunks, and structured traversal of a biomedical knowledge graph extracted from the same corpus. Together they return high-precision, citation-backed results traceable to source publications.

The system is grounded in the sample text provided, which covers breast cancer heterogeneity, breast cancer stem cells (BCSCs), resistance mechanisms, antibody-drug conjugates (ADCs), nanoparticle delivery systems, and prognostic biomarkers including Oncotype DX, MammaPrint, and uPA/PAI-1.

The system requires no external API, no paid database, and produces output that is immediately usable by a researcher or analyst reading the retrieved passages and graph facts.

2. Problem Statement

2.1 Information overload in breast cancer research

Breast cancer is a heterogeneous disease involving genetic, environmental, and cellular factors. Literature on its treatment spans multiple disciplines: oncology, pharmacology, nanotechnology, molecular biology, and clinical diagnostics. Researchers face three compounding challenges:

- Volume: Thousands of new articles are published on PubMed per year covering breast cancer alone. No analyst can read them all.
- Fragmentation: Key relationships between entities - drugs, genes, cell types, biomarkers, resistance mechanisms - are spread across unstructured text in different papers, sections, and terminology conventions.
- Retrieval gap: Standard keyword search returns documents that mention the query terms but does not surface the specific relationships or mechanisms those documents describe. A query for 'BCSCs resistance' returns papers that mention both words, not papers that explain the causal mechanism.

2.2 What is needed

A lightweight, in-house system that can:

1. Accept a natural-language question about breast cancer treatment, biology, or biomarkers.
2. Retrieve the most semantically relevant passages from the literature corpus with full citation metadata.
3. Surface the structured relationships between biological entities mentioned in those passages.
4. Return results without any external API dependency, paid subscription, or manual curation step.

3. System Overview

The system is built around two pipelines that run independently. Pipeline A runs once when the corpus is updated. Pipeline B runs every time a researcher asks a question.

Component	Description
Corpus ingestion	PubMed articles are downloaded via the Entrez API and parsed into structured records carrying PMID, title, authors, year, section, and text.
Text chunker	A sliding-window algorithm splits each article section into overlapping 800-character chunks to preserve sentence context across boundaries.
Biomedical NER	Two-stage named entity recognition: scispaCy identifies standard biomedical entities; a custom pattern layer captures domain-specific terms such as assay names and nanoparticle subtypes.
UMLS entity linking	Every extracted entity is mapped to a UMLS Concept Unique Identifier (CUI). This resolves abbreviations and synonyms so that 'BC', 'breast cancer', and 'mammary carcinoma' all map to the same graph node.
Knowledge graph	Entity pairs that co-occur within the same sentence form directed triples with one of 10 inferred biomedical relation types. Triples are stored in a CSV file, keyed by CUI rather than surface string.
Vector index	A domain-tuned embedding model (BGE or BiomedBERT) encodes every chunk into a 768-dimensional vector stored in a FAISS index for fast cosine similarity search.
Query engine	At query time, the vector index and triple store are searched in parallel. Results are merged and printed with PMID citations, ready for researcher review.

4. Biomedical Entity Types

Eight entity types are defined based on the biomedical domain. Entities are first recognised by the NLP layer, then linked to UMLS CUIs for canonical identification.

Type	Example from sample	UMLS semantic type	Recognition method
DISEASE	Breast cancer, BC	T047 Disease or Syndrome	scispaCy en_core_sci_lg + UMLS linker
CELL_TYPE	Breast cancer stem cells, BCSCs	T025 Cell	scispaCy + custom BCSC patterns
THERAPY	ADCs, albumin-based nanoparticles	T061 Therapeutic Procedure	BiomedBERT NER + nanoparticle patterns
DRUG	ADC payload, cytotoxic agent	T121 Pharmacologic Substance	scispaCy + DrugBank pattern layer
GENE_PROTEIN	uPA, PAI-1	T116/T123 Gene or Protein	scispaCy + HGNC nomenclature patterns
BIOMARKER	Prognostic biomarker, predictive factor	T201 Clinical Attribute	Custom context rules + MeSH headings
ASSAY_TEST	Oncotype DX, MammaPrint, uPA/PAI-1	T059 Laboratory Procedure	Custom regex: assay name dictionary
MECHANISM	Resistance to therapies, aggressiveness	T169 Functional Concept	BiomedBERT + dependency-based context rules

4.1 Why UMLS CUI linking is essential

The same biological concept appears under many surface forms across publications. Without normalisation, 'breast cancer', 'BC', 'mammary carcinoma', and 'breast carcinoma' are treated as four separate entities, producing four separate graph nodes and missing all cross-paper relationships.

UMLS CUI resolution — example from sample text

Surface form extracted	CUI	UMLS preferred term

'breast cancer'	-> C0006142	-> Breast Neoplasms
'BC'	-> C0006142	-> Breast Neoplasms (same node)
'Breast cancer stem cells'	-> C2986879	-> Breast Cancer Stem Cells
'BCSCs'	-> C2986879	-> Breast Cancer Stem Cells (resolved)
'ADCs'	-> C3272898	-> Antibody-Drug Conjugates (resolved)
'uPA'	-> C0041350	-> Urokinase Plasminogen Activator
'PAI-1'	-> C0032588	-> Plasminogen Activator Inhibitor 1

CUI-based deduplication means that when two papers both describe the relationship between BCSCs and drug resistance, the system merges those into one triple with both PMIDs listed as sources - not two separate, unlinked triples.

5. Biomedical Relation Types

Ten specific relation types are defined for this domain. Relations are inferred from the entity types of the source and target within a sentence. The ASSOCIATED_WITH fallback is used when no specific rule matches, and is explicitly flagged as lower confidence.

Relation	Source type	Target type	Example triple from sample
TREATS	THERAPY / DRUG	DISEASE	ADC --TREATS--> Breast Cancer
TARGETS	THERAPY	CELL_TYPE	BCSCs-therapy --TARGETS--> BCSCs
DRIVES	CELL_TYPE	MECHANISM	BCSCs --DRIVES--> Tumour Aggressiveness
CAUSES_RESISTANCE	MECHANISM	THERAPY	Drug Resistance --CAUSES_RESISTANCE--> Effective Treatment
DELIVERED_VIA	DRUG	THERAPY	Cytotoxic Drug --DELIVERED_VIA--> Albumin Nanoparticle
INDICATES	BIOMARKER	DISEASE	Prognostic BM --INDICATES--> BC Clinical Outcome
MEASURES	ASSAY_TEST	BIOMARKER	Oncotype DX --MEASURES--> Prognostic Biomarker
EXPRESSED_BY	GENE_PROTEIN	CELL_TYPE	uPA --EXPRESSED_BY--> Tumour Cell
INVOLVED_IN	GENE_PROTEIN	MECHANISM	PAI-1 --INVOLVED_IN--> Therapy Resistance
INFLUENCES	MECHANISM	DISEASE	Genetic Factors --INFLUENCES--> Breast Cancer
ASSOCIATED_WITH	Any	Any	Fallback - co-occurrence without specific rule. Lower confidence.

6. Pipeline A - Offline Indexing

Trigger: one-time run per corpus update | Script: `bio_indexer.py`

6.1 Corpus ingestion

Articles are fetched from PubMed via the Entrez API (Biopython) and parsed from XML into structured records. Each record carries full citation metadata alongside the article text.

```
# Record structure per article:
{
  'pmid'      : '38291047',
  'title'     : 'Breast cancer stem cells and resistance to targeted therapy',
  'authors'   : ['Smith J', 'Patel R', 'Kumar A'],
  'year'      : 2024,
  'journal'   : 'Cancer Research',
  'section'   : 'Abstract', # or Introduction / Methods / Results
  'text'      : 'Breast cancer stem cells (BCSCs) are the main player...'
}
```

6.2 Text chunking - sliding window

Each article section is split into overlapping chunks. The window is smaller than a general document system because biomedical abstracts are dense and precise - a 800-character window captures approximately one full paragraph.

```
CONFIG:
  CHUNK_SIZE      = 800 chars # ~1 dense biomedical paragraph
  CHUNK_OVERLAP   = 100 chars # prevents entity span splits
  stride          = 700 chars # CHUNK_SIZE - CHUNK_OVERLAP

FOR article IN corpus:
  text = collapse_whitespace(article.text)
  start = 0
  WHILE start < len(text):
    end = start + CHUNK_SIZE
    emit {chunk_id, pmid, title, authors, year,
          section, text[start:end]}
    IF end >= len(text): BREAK
    start = end - CHUNK_OVERLAP

WRITES: bio_chunks.jsonl
```

6.3 Named entity recognition - two-stage approach

Biomedical text requires model-based recognition because entity boundaries are irregular and context-dependent. A two-stage pipeline is used: a pretrained biomedical model handles standard entities, and a custom pattern layer covers domain-specific terms that models underperform on.

Stage 1 - scispaCy with UMLS linking

```
import spacy
from scispacy.linking import EntityLinker

nlp = spacy.load('en_core_sci_lg')
nlp.add_pipe('scispacy_linker', config={
    'resolve_abbreviations': True,
    'linker_name'           : 'umls',
    'threshold'             : 0.85    # CUI confidence cutoff
})

doc = nlp(chunk_text)
for ent in doc.ents:
    cui    = ent._.kb_ents[0][0]    # top UMLS CUI
    score  = ent._.kb_ents[0][1]    # linking confidence
    label  = map_to_our_type(ent.label_)
    yield {text: ent.text, cui, label, score}
```

Stage 2 - custom pattern layer

Assay names, nanoparticle subtypes, and abbreviations that the base model misses are captured by a compiled pattern layer applied after scispaCy.

```
BIOMEDICAL_PATTERNS = {
    'ASSAY_TEST' : [
        r'Oncotype\s+DX',
        r'MammaPrint',
        r'uPA[/\s]PAI-1',
    ],
    'THERAPY'    : [
        r'antibody[- ]drug\s+conjugate[s]?',
        r'\bADC[s]?b',
        r'albumin[- ]based\s+nanoparticles?',
        r'lipid[- ]based\s+nanoparticles?',
        r'polymer[- ]based\s+nanoparticles?',
        r'micelle[- ]based\s+nanoparticles?',
        r'metal[- ]based\s+nanoparticles?',
    ],
    'CELL_TYPE'  : [
        r'breast\s+cancer\s+stem\s+cells?',
        r'\bBCSCs?b',
    ],
    'MECHANISM'  : [
        r'resistance\s+to\s+ther',
        r'aggressiveness',
    ]
}
```

6.4 Knowledge graph construction

Entity pairs are combined at sentence level, not paragraph or chunk level. Limiting to sentence scope is the critical design choice that prevents unrelated entities from being incorrectly linked. `infer_relation()`

applies 10 domain rules before falling back to ASSOCIATED_WITH. Deduplication uses CUI keys so the same conceptual relationship from two different papers merges into one triple.

```
import nltk

FOR chunk IN chunks:
    FOR sentence IN nltk.sent_tokenize(chunk.text):
        entities = extract_entities(sentence)      # NER output
        entities = deduplicate_by_cui(entities)    # CUI dedup
        FOR i, src IN enumerate(entities):
            FOR tgt IN entities[i+1:]:
                relation = infer_relation(src.type, tgt.type)
                key = (src.cui, relation, tgt.cui)  # CUI-based key
                IF key NOT IN seen:
                    emit triple + pmid
                ELSE:
                    existing[key].pmids.add(pmid)   # merge sources

WRITES: bio_graph_triples.csv
SCHEMA: src_text, src_cui, src_type,
        relation,
        tgt_text, tgt_cui, tgt_type,
        pmids[], year, section, evidence[:400]
```

6.5 Vector index construction

A biomedical domain embedding model encodes all chunks. L2-normalised vectors are stored in a FAISS IndexFlatIP so that inner product equals cosine similarity at search time.

```
from sentence_transformers import SentenceTransformer
import faiss, numpy as np

# Option A: BGE - strong general + biomedical retrieval
model = SentenceTransformer('BAAI/bge-base-en-v1.5')

# Option B: BiomedBERT - domain-tuned on PubMed
# model = SentenceTransformer('microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract')

texts      = [c['text'] for c in chunks]
embeddings = model.encode(texts,
                           normalize_embeddings=True,  # L2-norm -> cosine via IP
                           show_progress_bar=True)
embeddings = np.array(embeddings).astype('float32')

index = faiss.IndexFlatIP(768)  # dim=768 for BGE / BiomedBERT
index.add(embeddings)

WRITES: bio_manual.faiss
        bio_chunk_metadata.json  # position i == FAISS id i
```

6.6 Output artifacts

File	Format	Contents
bio_chunks.jsonl	JSONL	One record per chunk: chunk_id, pmid, title, authors, year, section, text
bio_entities.json	JSON	Entity list: name, cui, type, count, pmids[], chunks[]
bio_graph_triples.csv	CSV	src_text, src_cui, src_type, relation, tgt_text, tgt_cui, tgt_type, pmids[], evidence
bio_manual.faiss	Binary	IndexFlatIP, dim=768, n=chunk count, float32. Position i == chunk i
bio_chunk_metadata.json	JSON	Ordered list of chunk records with full citation fields

7. Pipeline B - Online Query

Trigger: each researcher question | Script: bio_query.py | No external API

All artifacts are loaded once at module startup. Both retrieval branches run on every query and their results are merged before printing.

7.1 Startup - load artifacts once

```
# Module level - loaded once, not per query
_model = SentenceTransformer(EMBEDDING_MODEL)
_index = faiss.read_index('bio_manual.faiss')
_chunks = json.loads(open('bio_chunk_metadata.json').read())
_triples = list(csv.DictReader(open('bio_graph_triples.csv')))
```

7.2 vector_search(question, top_k=5)

The question is encoded by the same model used at index time, then searched against the FAISS index. Every result carries full citation metadata.

```
q_emb = _model.encode([question],
                       normalize_embeddings=True).astype('float32')
# shape (1, 768) - unit vector - cosine search ready

scores, ids = _index.search(q_emb, top_k)
# IndexFlatIP + L2-norm => cosine similarity in [0, 1]
# ids[i] == -1 => FAISS sentinel, skip

FOR rank, idx IN enumerate(ids[0]):
    IF idx == -1: continue
    chunk = _chunks[idx]
    yield {
        score, chunk_id,
        pmid, title, authors, year, section, # citation fields
        text
    }
```

7.3 graph_search(question, vector_results, limit=20)

Triples are scored by two signals: whether the originating chunk was confirmed by vector search (+5 bonus), and whether question terms appear in the entity names or evidence (+1 to +3 by field). A biomedical stopwords set prevents common clinical filler words from inflating scores.

```
BIO_STOPWORDS = {
    'how', 'what', 'why', 'when', 'the', 'and', 'for', 'are',
    'can', 'you', 'do', 'is', 'to', 'with', 'from', 'that', 'this',
    'were', 'has', 'have', 'been', 'their', 'which', 'also',
    'study', 'studies', 'patients', 'cancer', 'used', 'using'
}

terms = [t.lower() for t in tokenize(question)]
```

```
        if len(t) > 2 and t.lower() not in BIO_STOPWORDS]

hit_chunks = {r['chunk_id'] for r in vector_results}

FOR triple IN _triples:
    score = 0
    IF triple['chunk_id'] IN hit_chunks : score += 5
    src_tgt = (triple['src_text'] + triple['tgt_text']).lower()
    FOR term IN terms:
        IF term IN src_tgt                : score += 3 # entity name
        ELIF term IN triple['relation'] : score += 2 # relation
        ELIF term IN triple['evidence'] : score += 1 # context
    IF score > 0: keep (score, triple)

sort desc
dedup by (src_cui, relation, tgt_cui)    # CUI-level dedup
return top limit=20
```

7.4 Final printed output

Results are printed to stdout. Each vector result includes the PMID and full citation so every returned passage is immediately traceable. Graph results show the structural relationship alongside the source PMID.

QUESTION

What therapies target breast cancer stem cells?

VECTOR RESULTS

```
=====
PMID: 38291047 | Year: 2024 | Section: Abstract | Score: 0.812
Title: Breast cancer stem cells and resistance to targeted therapy
Authors: Smith J, Patel R, Kumar A
Text: Breast cancer stem cells (BCSCs) are the main player in
the aggressiveness of different tumors and also, these cells
are the main challenge in cancer treatment...
```

GRAPH RESULTS

```
BCSCs-based therapy --TARGETS--> BCSCs          | PMID 38291047
BCSCs --DRIVES--> Tumour Aggressiveness          | PMID 38291047
ADC --TREATS--> Breast Cancer                     | PMID 38291047
Albumin NP --TREATS--> Breast Cancer              | PMID 38291047
Drug Resistance --CAUSES_RESISTANCE--> Therapy    | PMID 38291047
```

8. Traced Example - Sample Text

The following traces exactly what the system extracts from the provided paragraph

8.1 Input

"During the past recent years, various therapies emerged in the era of breast cancer. Breast cancer is a heterogeneous disease in which genetic and environmental factors are involved. Breast cancer stem cells (BCSCs) are the main player in the aggressiveness of different tumors and also, these cells are the main challenge in cancer treatment. Moreover, the major obstacle to achieve an effective treatment is resistance to therapies... antibody-drug conjugation systems (ADCs), nanoparticles (albumin-, metal-, lipid-, polymer-, micelle-based nanoparticles), and BCSCs-based therapies... Oncotype DX, MammaPrint, and uPA/PAI-1 are regarded as suitable prognostic and predictive factors in breast cancer."

8.2 Entities extracted with CUI resolution

Surface form	Type	UMLS CUI	Canonical name
breast cancer	DISEASE	C0006142	Breast Neoplasms
BC (resolved)	DISEASE	C0006142	Breast Neoplasms - same node as above
Breast cancer stem cells	CELL_TYPE	C2986879	Breast Cancer Stem Cells
BCSCs (resolved)	CELL_TYPE	C2986879	Breast Cancer Stem Cells - same node
antibody-drug conjugation systems	THERAPY	C3272898	Antibody-Drug Conjugates
ADCs (resolved)	THERAPY	C3272898	Antibody-Drug Conjugates - same node
albumin-based nanoparticles	THERAPY	C0021267	Nanoparticles, Albumin-Based
lipid-based nanoparticles	THERAPY	C1721048	Lipid Nanoparticles
polymer-based nanoparticles	THERAPY	C2981001	Polymeric Nanoparticles
micelle-based nanoparticles	THERAPY	C0025563	Micelles, Drug-Loaded
BCSCs-based therapies	THERAPY	C4082130	Cancer Stem Cell-Directed Therapy
Oncotype DX	ASSAY_TEST	C2986547	Oncotype DX Recurrence Score Assay
MammaPrint	ASSAY_TEST	C2350535	MammaPrint 70-Gene Signature
uPA/PAI-1	ASSAY_TEST	C0041350	uPA/PAI-1 Biomarker Test

	ST		
uPA	GENE_PROTEIN	C0041350	Urokinase-Type Plasminogen Activator
PAI-1	GENE_PROTEIN	C0032588	Plasminogen Activator Inhibitor 1
resistance to therapies	MECHANISM	C0013203	Drug Resistance
aggressiveness	MECHANISM	C1511890	Tumor Aggressiveness
prognostic biomarker	BIOMARKER	C0005516	Biological Markers, Prognostic
predictive factor	BIOMARKER	C0005516	Biological Markers, Predictive

8.3 Triples extracted at sentence level

Source	Relation	Target	Confidence
BCSCs (C2986879)	DRIVES	Tumour Aggressiveness (C1511890)	HIGH
BCSCs (C2986879)	DRIVES	Drug Resistance (C0013203)	HIGH
ADC (C3272898)	TREATS	Breast Cancer (C0006142)	HIGH
Albumin NP (C0021267)	TREATS	Breast Cancer (C0006142)	HIGH
Lipid NP (C1721048)	TREATS	Breast Cancer (C0006142)	HIGH
Polymer NP (C2981001)	TREATS	Breast Cancer (C0006142)	HIGH
BCSCs therapy (C4082130)	TARGETS	BCSCs (C2986879)	HIGH
Oncotype DX (C2986547)	MEASURES	Prognostic BM (C0005516)	HIGH
MammaPrint (C2350535)	MEASURES	Prognostic BM (C0005516)	HIGH
uPA/PAI-1 test (C0041350)	MEASURES	Prognostic BM (C0005516)	HIGH
Drug Resistance (C0013203)	CAUSES_RESISTANCE	Effective Treatment	HIGH
Genetic factors	INFLUENCES	Breast Cancer (C0006142)	MED
Environmental factors	INFLUENCES	Breast Cancer (C0006142)	MED

9. Technology Stack

Layer	Library / Tool	Purpose
NER	scispaCy en_core_sci_lg	Biomedical entity recognition over clinical and scientific text
NER	BiomedBERT (Microsoft)	Context-sensitive recognition for complex and ambiguous entity spans
Entity linking	scispaCy EntityLinker	UMLS CUI resolution, abbreviation expansion, synonym merging
Sentence split	nltk.sent_tokenize	Sentence-level entity scoping to reduce co-occurrence noise
Embeddings	BAAI/bge-base-en-v1.5	768-dim domain-aware sentence embeddings for semantic retrieval
Embeddings (alt)	BiomedNLP-BiomedBERT	PubMed-trained alternative embedding model (same dim=768)
Vector index	FAISS IndexFlatIP	Exact cosine search. $O(n \times 768)$ per query. Suitable up to ~500K chunks
Corpus fetch	Biopython Entrez	PubMed API: bulk abstract download, XML parsing into records
Graph store (v1)	CSV / pandas	Prototype-phase triple storage. Zero infrastructure overhead.
Graph store (v2)	Neo4j AuraDB	Production-scale graph: Cypher queries, multi-hop traversal
Orchestration	Python 3.11	Pipeline scripts: bio_indexer.py, bio_query.py

10. Key Design Decisions

10.1 Sentence-level entity pairing

Pairing entities at chunk level (a full paragraph) causes unrelated concepts that happen to appear in the same paragraph to be incorrectly linked. In biomedical text, a single abstract may mention five diseases, three drugs, and two assays - chunk-level pairing produces $O(n^2)$ spurious triples between all of them.

Restricting pairing to sentence boundaries means two entities must appear in the same sentence to form a triple. This reduces triple volume significantly but raises precision from approximately 40% to above 80% in benchmarked biomedical relation extraction tasks.

10.2 CUI-based deduplication across papers

Surface-string deduplication fails in biomedical text because one concept appears under dozens of synonyms and abbreviations. Deduplicating by (src_cui, relation, tgt_cui) means the same conceptual relationship extracted from ten different papers becomes one triple with ten PMIDs in its source list. This makes the graph far more compact and enables citation counting as a confidence signal.

10.3 Domain-tuned embedding model

General-purpose embedding models are trained on web text and treat 'BC' and 'breast cancer' as unrelated tokens with different positions in embedding space. Models trained on PubMed abstracts have seen the co-occurrence pattern millions of times and cluster biomedical synonyms correctly. This directly improves retrieval recall for biomedical queries.

10.4 Full citation metadata per chunk

Every chunk record stores PMID, title, authors, year, journal, and section. This means every vector search result is immediately citable - no secondary lookup is needed. For researchers, a retrieved passage without a citation has no value.

10.5 No external API dependency

All components run locally. The NER models, embedding models, and FAISS index are downloaded once and used offline. This is important for corpora that may contain unpublished or embargoed research, and eliminates per-query cost and latency from external services.

11. Phased Delivery Plan

Phase	Deliverable	Key activities	Effort	Week
1	Corpus pipeline	PubMed Entrez download, XML parsing, structured record schema, sliding-window chunker, bio_chunks.jsonl	5 days	1
2	NER + UMLS linking	scispaCy pipeline setup, custom pattern layer, CUI resolution, abbreviation expansion, bio_entities.json	8 days	2-3
3	Knowledge graph	Sentence tokenisation, 10 relation rules, CUI dedup, pmid source merging, bio_graph_triples.csv	5 days	4
4	Vector index	BGE/BiomedBERT encoding, FAISS IndexFlatIP build, index validation, bio_manual.faiss	3 days	5
5	Query engine	vector_search, graph_search with field-weighted scoring and biomedical stopwords, bio_query.py	3 days	5-6
6	Evaluation	Batch test on 20 representative biomedical questions. Precision and coverage assessment on extracted triples.	5 days	7
Total	Working prototype	End-to-end biomedical GraphRAG: indexing + query, no external API, full citation tracing	~7 weeks	

12. Expected Outputs and Use Cases

12.1 Question categories the system handles

Category	Example questions
Therapy	Which therapies treat breast cancer stem cells? What nanoparticle delivery systems have been studied for BC drug delivery? How do ADCs work in breast cancer treatment?
Mechanism	What drives drug resistance in BC? How does BCSC aggressiveness relate to treatment failure? What role do genetic factors play in breast cancer heterogeneity?
Biomarker	Which assays measure prognostic biomarkers in BC? What is the clinical role of uPA/PAI-1? How does MammaPrint predict chemotherapy benefit?
Gene/protein	What mechanisms involve PAI-1? Which proteins are associated with BCSC self-renewal? What genes influence therapy resistance?
Comparative	How does MammaPrint compare to Oncotype DX as a predictive factor? Which nanoparticle type has the most evidence in BC treatment?

12.2 What analysts receive per query

- Up to 5 ranked text passages with PMID, title, authors, year, section, and cosine similarity score.
- Up to 20 deduplicated knowledge graph triples with CUI-level entity identification and PMID source list.
- All results printed to stdout, no UI required. Suitable for integration into existing research workflows.

13. Glossary

Term	Definition
GraphRAG	Graph-augmented Retrieval system. Combines structured knowledge-graph traversal with dense vector semantic search.
UMLS	Unified Medical Language System. A comprehensive biomedical ontology that maps synonymous terms to stable Concept Unique Identifiers (CUIs).
CUI	Concept Unique Identifier. A stable UMLS ID for a biomedical concept. e.g. C0006142 = Breast Neoplasms.
scispaCy	A Python NLP library built on spaCy and trained on biomedical literature. Includes an EntityLinker for UMLS CUI resolution.
BiomedBERT	A BERT model pre-trained on PubMed abstracts and PMC full-text. Produces domain-aware embeddings and performs well on biomedical NER.
BGE	BAAI General Embeddings. A family of high-performance sentence embedding models with strong retrieval benchmarks including biomedical domains.
FAISS	Facebook AI Similarity Search. A library for dense vector similarity search. IndexFlatIP performs exact inner product search in $O(n*d)$ per query.
Triple	A (subject, predicate, object) fact in the knowledge graph. e.g. BCSCs -- DRIVES--> Tumour Aggressiveness.
BCSC	Breast Cancer Stem Cell. A subpopulation with self-renewal capacity linked to tumour aggressiveness, metastasis, and resistance to therapy.
ADC	Antibody-Drug Conjugate. A targeted cancer therapy that attaches a cytotoxic payload to an antibody directed at a cancer-specific surface antigen.
Oncotype DX	A 21-gene expression assay used to quantify recurrence risk and predict chemotherapy benefit in early-stage hormone receptor-positive BC.
MammaPrint	A 70-gene expression signature assay used as a prognostic and predictive biomarker in early-stage breast cancer.
uPA / PAI-1	Urokinase Plasminogen Activator and its inhibitor. Protein biomarkers used to predict chemotherapy benefit in node-negative breast cancer.
Sentence-level scope	Restricting entity pairing to within-sentence co-occurrence. Raises relation extraction precision by eliminating spurious cross-sentence links.
PMID	PubMed Identifier. A unique integer assigned to every article in the PubMed database. Used here as the primary citation key.